

BANARAS HINDU UNIVERSITY

B.A. (Hons.) — Semester IV Examination, 2023-24

Paper No. MTB-402: Mathematical Methods

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 076466

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) Define Laplace transform. Evaluate $\mathcal{L}\{e^t\}$ for all $t > 0$.
- (b) Define a piecewise continuous function. Find the Laplace transform of a piecewise continuous function.
- (c) What is a function of exponential order? Give an example of a function that is NOT of exponential order and prove it.
- (d) Evaluate $\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\}$.
- (e) Find $\mathcal{L}\{t(3\cos 2t - 2\sin 2t)\}$.
- (f) Find the Laplace transform of $\int_0^t \sin 2u \, du$.
- (g) Explain the change-of-scale property of the Laplace transform.
- (h) Differentiate between the initial and final value theorems.
- (i) Find the Laplace transform of the cosine integral.
- (j) Give an example of an odd function and expand it in a sine series.
- (k) Define a functional. Find the shortest smooth plane curve joining $P(x_1, y_1)$ and $Q(x_2, y_2)$.

Question 2.

- (a) Prove linearity of the Laplace transform. Hence evaluate $\mathcal{L}\{\cos^2 t \cos 3t\}$. [6]
- (b) If $f(t)$ is piecewise continuous on $[0, \infty)$ and of exponential order, prove $\mathcal{L}\{f(t)\}$ exists. [6]

Question 3.

- (a) Find $\mathcal{L}\{\sin \sqrt{t}\}$ using the series expansion method. [6]
- (b) If $F(t)$, $f(t)$ are continuous of exponential order and $f'(t)$ is at least piecewise continuous, prove $\mathcal{L}\{f''(t)\} = s^2\mathcal{L}\{f(t)\} - sf(0) - f'(0)$. [6]
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Question 4.

- (a) Solve by Laplace transform: $y'' - 3y' + 2y = e^{-t}$, $y(0) = 1$, $y'(0) = 4$. [6]
- (b) Prove $\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$ for every natural number n , where $\mathcal{L}\{f(t)\} = F(s)$. [6]
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Question 5.

- (a) State and prove the first shifting (translation) theorem. Hence evaluate $\mathcal{L}\{t^n e^{at}\}$. [6]
- (b) If $f(t)$ is periodic with period a , derive its Laplace transform. [6]
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Question 6.

- (a) Define a Fourier series and explain its convergence at points of continuity and discontinuity with examples. [6]
- (b) Expand $f(x) = x^2$, $0 < x < L$, in cosine, sine and full Fourier series. Explain the reflections given by each. [6]
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Question 7.

- (a) State and prove the Euler-Lagrange equation to obtain the extremum of a functional. [6]
- (b) Find the extremal for $I[y] = \int_0^\pi [y'^2 - y^2 - 2y \cos(\frac{\pi}{2} - x)] dx$, $y(0) = 0$, $y(\pi) = 1$. [6]
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